CIE International A Level Physics

# Topic 19

# Capacitance

Revision Booklet

**This booklet covers:**

- Capacitors and Capacitance
- Capacitors in Series and Parallel
- Energy Stored in a Capacitor
- Capacitor Discharge
- The Time Constant

## Core Concepts and Definitions

### Capacitance

The **capacitance** of a conductor is the charge stored per unit potential difference across it.

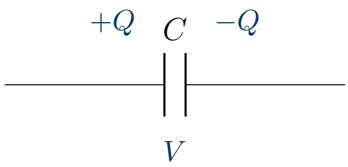
$$C = \frac{Q}{V} \qquad \text{units: F (farads)} \equiv \text{C V}^{-1}$$

- 1 F is a very large unit; practical capacitors are typically $\mu$F, nF or pF.
- Capacitance depends on the **geometry** of the conductor and the medium between the plates, not on $Q$ or $V$ individually.
- For an **isolated spherical conductor** of radius $R$: $C = 4\pi\varepsilon_0 R$.

### The Parallel Plate Capacitor

Two conducting plates of area $A$ separated by distance $d$:

- Charging the capacitor stores charge $+Q$ on one plate and $-Q$ on the other.
- The electric field between the plates is uniform: $E = V/d$.
- Capacitance increases with larger plate area $A$ and smaller separation $d$.




## Capacitors in Series and Parallel

### Capacitors in Parallel

$$C_{\text{total}} = C_1 + C_2 + C_3 + \cdots$$

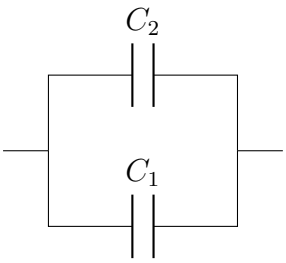
Each capacitor has the **same voltage**; charges add: $Q_{\text{total}} = Q_1 + Q_2 + \cdots$

### Capacitors in Series

$$\frac{1}{C_{\text{total}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots$$

Each capacitor has the **same charge**; voltages add: $V_{\text{total}} = V_1 + V_2 + \cdots$

### Circuit diagrams




**Parallel:** $C_T = C_1 + C_2$

**Series:** $\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2}$

**Derivation of Series Formula**

For capacitors in series, each carries the same charge $Q$. The total voltage is:

$$V_{\text{total}} = V_1 + V_2 = \frac{Q}{C_1} + \frac{Q}{C_2} = Q\left(\frac{1}{C_1} + \frac{1}{C_2}\right)$$

Since $C_{\text{total}} = Q/V_{\text{total}}$, dividing through by $Q$ gives $\dfrac{1}{C_{\text{total}}} = \dfrac{1}{C_1} + \dfrac{1}{C_2}$.

**Common Mistake**

Capacitors in series combine like **resistors in parallel** (reciprocal rule), and capacitors in parallel combine like **resistors in series** (direct sum). This is the opposite to resistors — don't mix them up.

## Energy Stored in a Capacitor

**Energy Stored**

$$W = \tfrac{1}{2}QV = \tfrac{1}{2}CV^2 = \frac{Q^2}{2C}$$

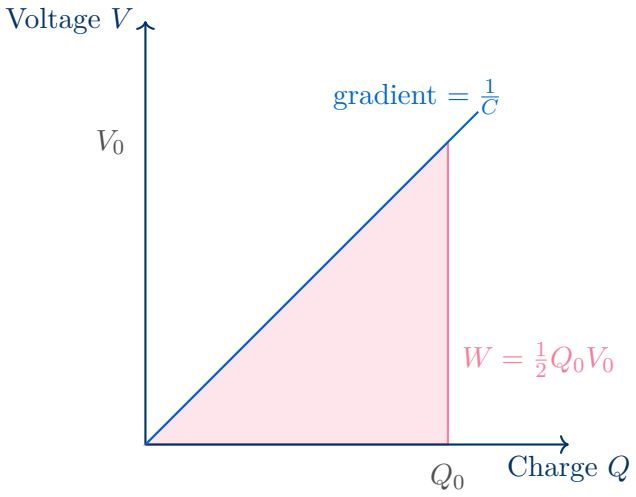
$W$ = energy stored (J)
$Q$ = charge stored (C)
$V$ = potential difference across capacitor (V)
$C$ = capacitance (F)

**Energy from the $V$–$Q$ Graph**

The energy stored equals the **area under the $V$–$Q$ graph** (a straight line through the origin with gradient $1/C$):

$$W = \text{area of triangle} = \tfrac{1}{2} \times Q \times V = \tfrac{1}{2}QV$$

### $V$–$Q$ graph for a capacitor




## Discharging a Capacitor

### Capacitor Discharge Through a Resistor

When a charged capacitor discharges through a resistor $R$, the charge, voltage and current all decay **exponentially** with time:

$$Q = Q_0\, e^{-t/RC} \qquad V = V_0\, e^{-t/RC} \qquad I = I_0\, e^{-t/RC}$$

where $I_0 = V_0/R = Q_0/RC$ is the initial current.

### Time Constant

$$\tau = RC$$

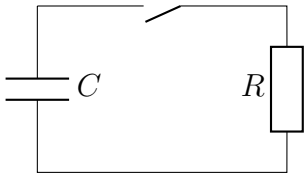
$\tau$ = time constant (s)
$R$ = resistance of discharge path ($\Omega$)
$C$ = capacitance (F)

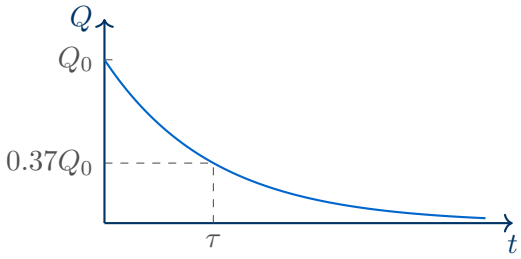
After one time constant ($t = \tau$), the charge/voltage/current has fallen to $e^{-1} \approx$ **37**% of its initial value.

### Key Values During Discharge

- $t = \tau$: $Q = 0.37\, Q_0$ (37%)
- $t = 2\tau$: $Q = 0.135\, Q_0$ (13.5%)
- $t = 5\tau$: $Q \approx 0.007\, Q_0$ — capacitor considered fully discharged.
- A larger $\tau$ means slower discharge; a smaller $\tau$ means faster discharge.

### Discharge circuit and $Q$–$t$ graph




**Analysing Discharge Graphs**

- A **linear** $\ln Q$ vs $t$ graph confirms exponential decay; gradient $= -1/RC$.
- The time constant $\tau$ can be read directly as the time for $Q$ to fall to $0.37\,Q_0$.
- Doubling $R$ or $C$ doubles $\tau$ and halves the rate of discharge.

### Linearising the discharge: $\ln Q$ against $t$

Taking logarithms of $Q = Q_0\,e^{-t/RC}$:

$$\ln Q = \ln Q_0 - \frac{1}{RC}\,t$$

This is of the form $y = mx + c$, so a graph of $\ln Q$ against $t$ gives:

- gradient $= -\dfrac{1}{RC} = -\dfrac{1}{\tau}$
- $y$-intercept $= \ln Q_0$

## Formula Summary Sheet

| Formula | Quantity | Units |
|---|---|---|
| $C = Q/V$ | Capacitance (definition) | F |
| $C_T = C_1 + C_2 + \cdots$ | Capacitors in parallel | F |
| $\dfrac{1}{C_T} = \dfrac{1}{C_1} + \dfrac{1}{C_2} + \cdots$ | Capacitors in series | F |
| $W = \frac{1}{2}QV = \frac{1}{2}CV^2 = \dfrac{Q^2}{2C}$ | Energy stored | J |
| $\tau = RC$ | Time constant | s |
| $Q = Q_0\, e^{-t/RC}$ | Charge during discharge | C |
| $V = V_0\, e^{-t/RC}$ | Voltage during discharge | V |
| $I = I_0\, e^{-t/RC}$ | Current during discharge | A |
| $\ln Q = \ln Q_0 - \dfrac{t}{RC}$ | Linearised discharge | — |

**Constants and values:** $e^{-1} \approx 0.368$; after $1\tau$: 37% remains; after $5\tau$: $< 1\%$ remains

**Units check:** $[\tau] = [\Omega][\mathrm{F}] = [\mathrm{V\ A^{-1}}][\mathrm{C\ V^{-1}}] = [\mathrm{C\ A^{-1}}] = \mathrm{s}$

## Worked Examples

### Example 1 — Capacitors in Series and Parallel

**Question:** Three capacitors of 2.0 $\mu$F, 3.0 $\mu$F and 6.0 $\mu$F are connected (a) in parallel and (b) in series. Find the total capacitance in each case.

**Solution**

**(a) Parallel:**

$$C_T = 2.0 + 3.0 + 6.0 = \mathbf{11.0}\ \mu\mathrm{F}$$

**(b) Series:**

$$\frac{1}{C_T} = \frac{1}{2.0} + \frac{1}{3.0} + \frac{1}{6.0} = \frac{3+2+1}{6.0} = \frac{6}{6.0} = 1.0\ \mu\mathrm{F}^{-1}$$

$$C_T = \mathbf{1.0}\ \mu\mathrm{F}$$

### Example 2 — Energy Stored

**Question:** A 470 $\mu$F capacitor is charged to 12 V. Calculate (a) the charge stored and (b) the energy stored.

**Solution**

**(a)** $Q = CV = 470 \times 10^{-6} \times 12 = \mathbf{5.64 \times 10^{-3}}$ C
**(b)** $W = \frac{1}{2}CV^2 = \frac{1}{2} \times 470 \times 10^{-6} \times 12^2 = \mathbf{3.38 \times 10^{-2}}$ J

### Example 3 — Capacitor Discharge

**Question:** A 220 $\mu$F capacitor charged to 9.0 V discharges through a 47 k$\Omega$ resistor. Calculate (a) the time constant and (b) the voltage after 5.0 s.

**Solution**

**(a)** $\tau = RC = 47 \times 10^3 \times 220 \times 10^{-6} = \mathbf{10.3}$ s
**(b)** $V = V_0\, e^{-t/RC} = 9.0 \times e^{-5.0/10.3} = 9.0 \times e^{-0.485} = 9.0 \times 0.616 = \mathbf{5.5}$ V

## Practice Exam Questions

### Section A — Short Answer Questions

**Q1.** Define capacitance and state its SI unit.

*[2 marks]*

**Q2.** A capacitor stores 360 $\mu$C of charge when connected to a 12 V supply. Calculate its capacitance.

*[2 marks]*

**Q3.** Two capacitors of 4.0 $\mu$F and 12 $\mu$F are connected in series across a 6.0 V supply.

(a) Calculate the total capacitance. *[2 marks]*

(b) Calculate the total charge stored. *[1 mark]*

(c) Calculate the voltage across the 4.0 $\mu$F capacitor. *[2 marks]*

**Q4.** Explain, with reference to the $V$–$Q$ graph, why the energy stored in a capacitor is $W = \frac{1}{2}QV$ and not $W = QV$.

*[2 marks]*

**Q5.** Define the time constant for a capacitor–resistor discharge circuit and state what fraction of the initial charge remains after two time constants.

*[3 marks]*

## Section B — Longer Structured Questions

**Q6.** A 100 $\mu$F capacitor is charged to 20 V and then discharged through a 25 k$\Omega$ resistor.

(a) Calculate the initial charge stored on the capacitor.

*[1 mark]*

(b) Calculate the initial discharge current.

*[2 marks]*

(c) Calculate the time constant for the discharge.

*[1 mark]*

(d) Calculate the charge remaining after 4.0 s.

*[2 marks]*

(e) The student plots a graph of $\ln(Q/\text{C})$ against $t/\text{s}$. State the gradient and $y$-intercept of this graph.

*[2 marks]*

**Q7.** A 50 $\mu$F and a 200 $\mu$F capacitor are connected in series and charged from a 15 V supply.

(a) Calculate the combined capacitance.

*[2 marks]*

(b) Calculate the energy stored in the combination.

*[2 marks]*

(c) The two capacitors are now reconnected in parallel across the same supply. Calculate the new total energy stored and explain why it differs from part (b).

*[3 marks]*

## Mark Scheme and Answers

**Q1.** Capacitance is the charge stored per unit potential difference [1]; unit: farad (F) or C V$^{-1}$ [1].

**Q2.** $C = Q/V = 360 \times 10^{-6}/12 = \mathbf{30}$ $\mu$F [2].

**Q3(a).** $\dfrac{1}{C_T} = \dfrac{1}{4.0} + \dfrac{1}{12} = \dfrac{3+1}{12} = \dfrac{4}{12}$ [1]; $C_T = \mathbf{3.0}$ $\mu$F [1].

**Q3(b).** $Q = C_T V = 3.0 \times 10^{-6} \times 6.0 = \mathbf{1.8 \times 10^{-5}}$ C [1].

**Q3(c).** Same charge on each capacitor in series; $V_1 = Q/C_1 = 1.8 \times 10^{-5}/(4.0 \times 10^{-6}) =$ **4.5** V [2].

**Q4.** The $V$–$Q$ graph is a straight line through the origin [1]; the energy is the area under this graph (a triangle), which is $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2}QV$ — not $QV$ because the voltage builds up gradually from 0 to $V$ as charge is stored [1].

**Q5.** The time constant is the product $RC$ [1]; it is the time taken for the charge (or voltage or current) to fall to $1/e \approx 37\%$ of its initial value [1]; after $2\tau$: $e^{-2} \approx \mathbf{13.5}\%$ remains [1].

**Q6(a).** $Q_0 = CV_0 = 100 \times 10^{-6} \times 20 = \mathbf{2.0 \times 10^{-3}}$ C [1].

**Q6(b).** $I_0 = V_0/R = 20/(25 \times 10^3)$ [1] $= \mathbf{8.0 \times 10^{-4}}$ A [1].

**Q6(c).** $\tau = RC = 25 \times 10^3 \times 100 \times 10^{-6} = \mathbf{2.5}$ s [1].

**Q6(d).** $Q = Q_0\, e^{-t/RC} = 2.0{\times}10^{-3}{\times}e^{-4.0/2.5}$ [1] $= 2.0{\times}10^{-3}{\times}e^{-1.6} = 2.0{\times}10^{-3}{\times}0.202 = \mathbf{4.0 \times 10^{-4}}$ C [1].

**Q6(e).** Gradient $= -1/RC = -1/2.5 = \mathbf{-0.40}$ $\mathbf{s^{-1}}$ [1]; $y$-intercept $= \ln Q_0 = \ln(2.0 \times 10^{-3}) = \mathbf{-6.2}$ [1].

**Q7(a).** $\dfrac{1}{C_T} = \dfrac{1}{50} + \dfrac{1}{200} = \dfrac{4+1}{200} = \dfrac{5}{200}$ [1]; $C_T = \mathbf{40}$ $\mu$F [1].

**Q7(b).** $W = \frac{1}{2}C_T V^2 = \frac{1}{2} \times 40 \times 10^{-6} \times 15^2$ [1] $= \mathbf{4.5 \times 10^{-3}}$ J [1].

**Q7(c).** $C_T = 50 + 200 = 250$ $\mu$F; $W = \frac{1}{2} \times 250 \times 10^{-6} \times 15^2$ [1] $= \mathbf{2.8 \times 10^{-2}}$ J [1]; more energy is stored in parallel because the total capacitance is greater — more charge is drawn from the supply [1].

## Revision Checklist

Use this checklist to track your understanding. Tick each box when you are confident:

| | Learning Objective | Confidence (1–3) |
|---|---|---|
| ☐ | Define capacitance; use $C = Q/V$ | |
| ☐ | Calculate combined capacitance for series and parallel combinations | |
| ☐ | Derive the series and parallel formulae from $C = Q/V$ | |
| ☐ | Use $W = \frac{1}{2}QV = \frac{1}{2}CV^2 = Q^2/2C$ for energy stored | |
| ☐ | Explain why energy stored is the area under a $V$–$Q$ graph | |
| ☐ | Describe the exponential decay of $Q$, $V$ and $I$ during discharge | |
| ☐ | Use $x = x_0\, e^{-t/RC}$ for discharge of charge, voltage or current | |
| ☐ | Define the time constant $\tau = RC$ and state its physical significance | |
| ☐ | Determine $\tau$ from a discharge graph (directly or via $\ln Q$ vs $t$) | |
| ☐ | Sketch discharge curves for $Q$, $V$ and $I$ against time | |
| ☐ | Linearise the discharge equation and interpret gradient and intercept | |

*Key: 1 = Need more work 2 = Getting there 3 = Confident*

**Good luck with your revision!**

The exponential decay of a capacitor is one of the most important mathematical forms in physics. Once you can recognise it, linearise it, and extract $\tau$ from a graph, you have a powerful tool that reappears throughout the course.